Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Numerical Methods

BCA — Semester IV — 2024 — Answer Sheet

Subject Code: CACS252

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group CAttempt any TWO questions[2x10=20]

2. Explain absolute and relative error. Find the relative error of number 5.6 if both of its digits are correct.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. On what type of equations Newton's methods can be applicable. Justify.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Solve the following equations by using Gauss-Jordan method.2x+3y+4z=5 3x+4y+5z=6 4x+5y+6z=7

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Use the Romberg method to get an improved estimate of the integral from x = 1.8 to x = 3.4 from the data in the table with h = 0.4.X :1.61.82.02.22.42.62.83.03.23.43.63.8Y4.956.057.389.0211.0213.4616.4420.0524.5329.9636.5944.70:309534563481

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write a program to compute integral0∫x/2√sinx dx Simpson's 1/3 rule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Using Runge-Kutta method of 4th order solve the following equation taking each step h = 0.1dydx=4xy- xy given y(0) = 3.3 calculate y at x = 0.1 and 0.2.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Solve the laplace equation Uxx + Uyy = 0 for the following square mesh with the boundary values.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Solve the given set of linear equations using Dolittle LU decomposition method:3x1+ 2x2+ x3= 102x1+ 3x2+ 2x3= 143x1+ 2x2+ 3x3= 14

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define initial value problems and final value problems. Using heun's method, find value of y when x=0.3 given thatdydx= x + y and y=1 when x=0.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How can x we use Laterpolation techniques (methods) to approximate the value of the integral for the functions whose antiderivative can't be found? Explain. Write a program to solve sin x - 2x + 1 = 0 using Bisection method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.